



PROFESSIONAL
EDUCATION



Digital Plus⁺
Programs

Digital Platforms:

Designing Two-Sided
Markets from APIs to
Feature Roadmaps



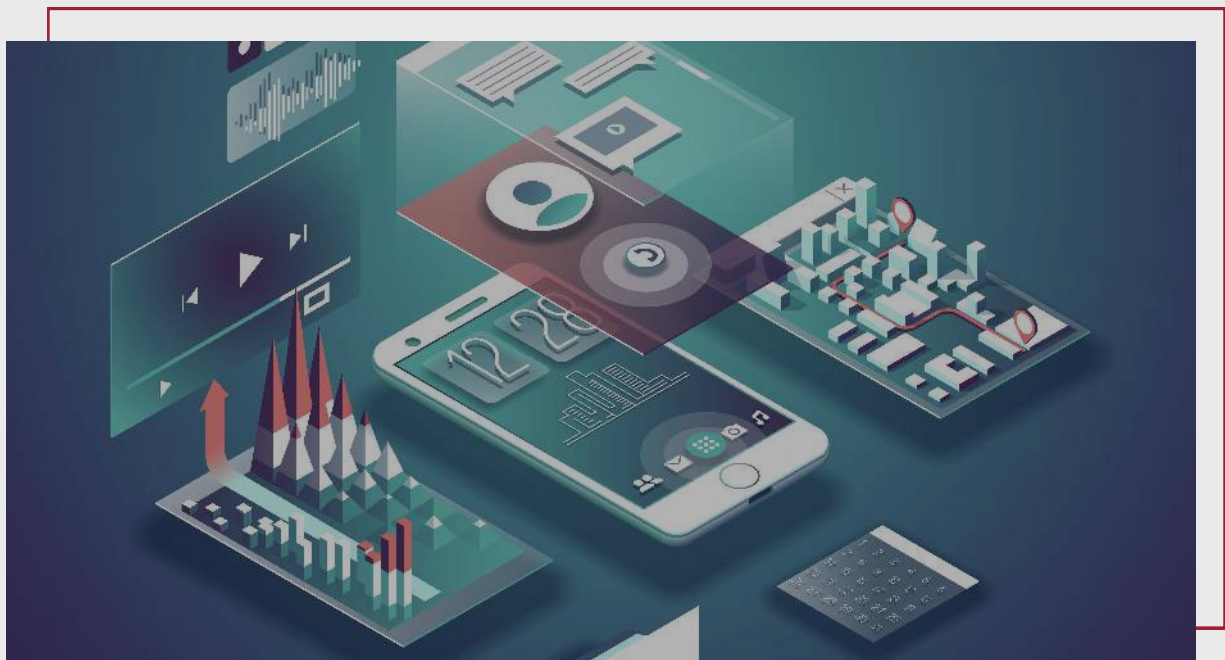
Digital platforms are everywhere (Uber, Airbnb, Android...), but a minority of platform attempts succeed: find out how to build a platform and the ecosystem around it.

Dr. Bruce Cameron

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Digital Platforms are Everywhere, but What Makes Them Tick?



Netflix, Uber, Facebook, Amazon, Apple: many of today's most wellknown organizations have found success through the implementation of digital platforms, multi-sided online markets that **effectively connect businesses, products, services, and information with customers** in beneficial new ways. And these modern platforms can be found in almost every industry, from the **service industry and social media to transportation, real estate, and software.**

Their vast growth in recent years, however, does not automatically guarantee business success. While many organizations are able to harness the capabilities of digital platforms to expand around the globe—securing funding and broadening their user bases—, many others fail to make an impact. So, what are businesses doing to ensure success?

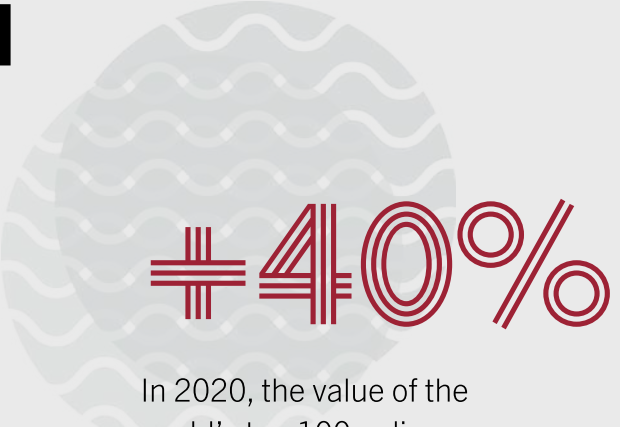
Digital Platforms: A Growing Business Model



1/5

The five largest platform companies (Alphabet, Amazon, Apple, Microsoft, and Facebook) [...] make up almost a fifth of the value of the S&P 500 Index.

Source: The Economist



+40%

In 2020, the value of the world's top 100 online platforms increased by 40%.*

Source: European Commission



X2

The sales growth of 43 innovation and transaction platforms from 1995-2015 was DOUBLE that of their non-platform peers.

Source: The Business Of Platforms: Strategy in the Age of Digital Competition, Innovation, and Power.

Decision Marking

Despite their reputation for rapidly expanding customer bases, **the long-term success of digital platform models depends more on how they're implemented.** Namely, sustained impact is often derived from the technical expertise and effective decision-making processes of managers and technical business leaders looking to grow their businesses in a customer-centric way.

Organizations are looking to leverage digital platforms to grow a loyal user base and drive customer demand for their products and services. With that in mind, an extensive understanding of the technical intricacies of digital platforms can offer an array of professionals—including managers, engineers, and entrepreneurs—a substantial competitive advantage.



About this Course



Digital platforms and two-sided markets have quickly established themselves as a **staple of modern business**, presenting new opportunities and challenges alike for engineers, product managers, regulators, and entrepreneurs.

However, there continues to be little discussion surrounding why some digital platforms help companies flourish where others have fallen short.

Participants will gain **real-world insights** into the development and scaling of digital platforms by hearing firsthand from professionals from a number of organizations, including **DoorDash, BlaBlaCar, LinkedIn, Etsy, and Uber Eats**.

MIT Professional Education's online course **Digital Platforms** aims to guide participants as they **design and develop effective digital platforms**.

This course employs a **technical approach to building loyal consumer bases and differentiated products**, taking an in-depth look at features, APIs and standards.



8 weeks



8-10 hours a week



Online

Key Takeaways

This course is designed to help participants build their own successful digital platforms.

It focuses on **technical and design aspects**, as well as on the importance of the **effective decision-making processes** that participants will encounter when developing a digital platform. These types of decisions may include when to **open a platform**, how to use **APIs to build an ecosystem of partners**, how to **shape an architecture** in an open environment, and how to find **new customers** in a digital platform.

The course is designed with practicality in mind and includes several **case studies** of successes and failures in modern markets.

Throughout this course, participants will:

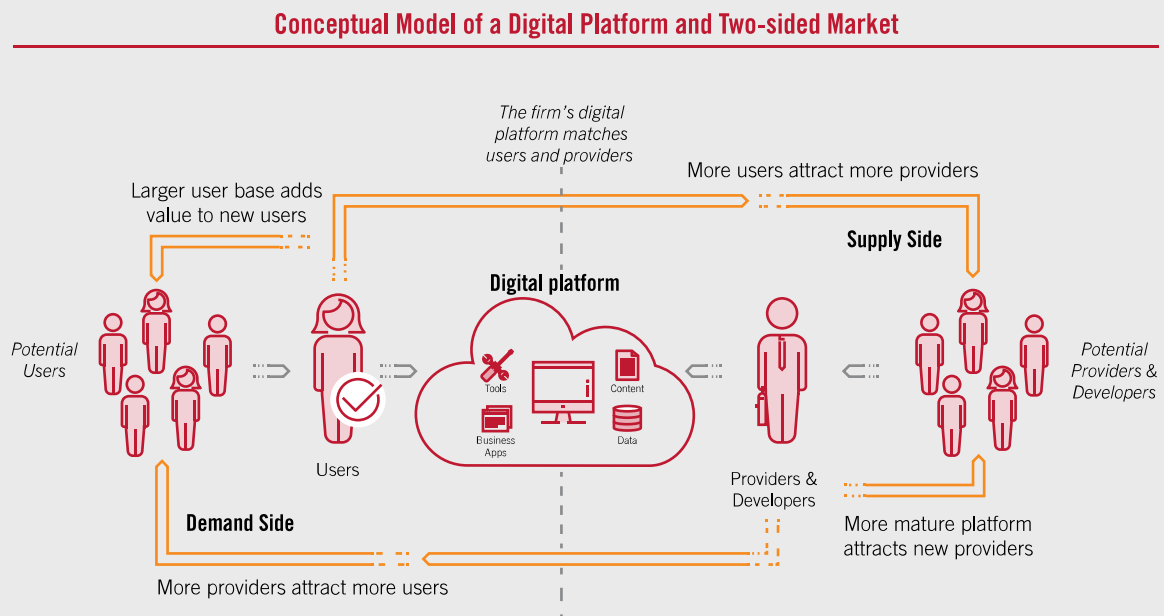
- 01 Recognize two-sided markets** and distinguish them from traditional onesided or direct buyer-seller markets.
- 02 Critically analyze and discuss examples of industry platforms** such as eBay, Etsy, Apple, Amazon, and Alibaba, and determine factors for success and failure.
- 03 Identify the technical choices a firm makes** in creating APIs and industry standards for two-sided markets enabled by digital platforms.
- 04 Plan for development and rollout** of a two-sided market, including adding features, incentivizing key customers, and negotiating with partners.
- 05 Examine the most recent research and models** on two-sided markets, including on the stability of two-sided markets.
- 06 Define metrics for success** in adoption and customer engagement.
- 07 Frame the tactics for growing provider and customer bases simultaneously** versus the firm's balance sheet.



Structure of a Typical Digital Platform and Two-sided Market

The demand side on the left side shows potential users, who may become users when they join the digital platform to purchase goods or services over the platform. The supply side on the right side consists of potential providers who become actual providers once they join the platform.

Transactions are mediated over the digital platform which has an owner who monetizes the volume of transactions and the current user and provider base. The increase (or decrease) in users and adopters is mediated by a number of factors such as network effects, as well as the value, quality and price of the goods and services offered over the platform



Participant Profile

This course is designed for professionals that operate digital platforms or are creating a business around a digital platform and are looking to leverage these platforms to make informed decisions, differentiate their products, and grow their customer base.

The contents can be of great benefit for professionals from a number of sectors and different levels, such as:

C-suite and other executives looking to gain a deeper technical understanding of digital platforms in order to make more effective decisions for their organization.

Developers and engineers who work with digital platforms and want to learn about the key technical choices of two-sided markets, such as software development kits or application courseming interfaces (APIs).

Product managers who want to leverage their technical choices for a dynamic digital platform strategy in the online marketplace.

Entrepreneurs looking to develop startups centered around digital platforms and two-sided markets and who want to ensure their company's success in the online marketplace.

Policy experts and regulator who work with digital market policies in an expanding global economy.

Professionals from technical backgrounds who work with digital platforms and two-sided markets who want to understand the choices they can make to enable their success.



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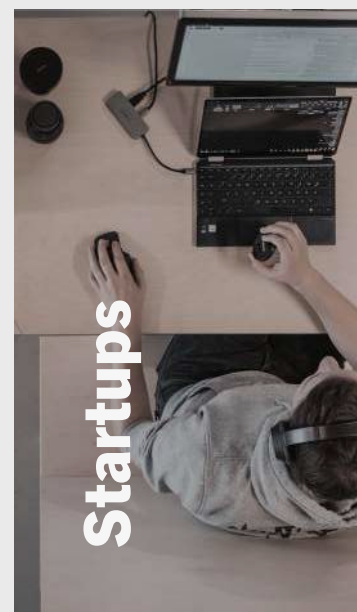
Real-World Applications



The high consumer demand for two-sided markets and online marketplaces means that the **demand for engineers and developers** who can build and operate digital platforms will increase as well. These professionals should be equipped with the **latest technical knowledge** and understand which technical choices can greatly increase the possibilities for business success.

With the rise and proliferation of digital platforms and two-sided markets, an increasing number of entrepreneurs are **looking to launch startups** based on digital platforms.

However, increasing competition, insufficient technical knowledge, and confusion regarding the dynamics of two-sided markets and online marketplaces could be fatal for their business operations. This course offers entrepreneurs the opportunity to acquire the expertise they need **to gain a strategic advantage** as they plan for and develop a new business.





The rapid expansion of digital platforms and two-sided markets means that **quick, effective policies must be developed** to regulate them.

In order for these policies to work, policymakers must first gain a broad understanding of the unique **intricacies of two-sided markets**.

An increasing number of customers are turning to digital platforms when **making online purchases** because of the variety, simplicity, and the efficient services they offer. Being a part of a digital platform is an excellent step that product and service providers can make as they look to **adapt and thrive in today's competitive industry**.



Instructors



Dr. Bruce Cameron

Director del MIT System Architecture Group

Dr. Bruce Cameron is director of the System Architecture Lab at MIT.

His areas of research include technology strategy, system architecture and product platform management.

He is currently the Faculty Director of MIT's online certificate "Architecture and System Engineering" and is working on various research projects: Analyzing the costs of supplier replacement and retention incentives in driven vehicles for hire (ride hailing), the effects of increased autonomy in commercial trucking on fleet management, and the effects of one-pilot operations on commercial flights.

"Digital platforms are some of the most interesting (and hardest!) technical challenges to work on today."

Dr. Cameron has led research projects for Amazon, Chevron, Sikorsky, Nokia, Caterpillar, and NASA.

His work in Sloan Executive Education has been rated as the highest level of Executive Education program at MIT for several years

He is a co-founder of Technology Strategy Partners and was the consultant for twelve companies in initial phases of online commerce platforms.



Prof. Olivier de Weck

Professor, Aeronautics and Astronautics and Engineering Systems, MIT

Olivier De Weck is a leader in systems engineering research. He focuses on how complex man-made systems such as aircraft, spacecraft, automobiles, printers, and critical infrastructures are designed and how they evolve over time. In addition, he studies strategic properties that have the potential to maximize lifecycle value and reduce programmed obsolescence.

Since 2001, his group has developed novel quantitative methods and tools that explicitly consider manufacturability, flexibility, commonality, and sustainability, among other characteristics. In his research, de Weck emphasizes excellence, innovation, and combining theory and practice.

“Developers of two-sided market platforms must acquire enough users on both sides in appropriate proportions to generate value to either side of the user market and thereby provide the foundation for sustained future growth.”

De Weck holds a degree in industrial engineering from ETH Zurich and an M.S. in aeronautics and astronautics from MIT

He earned his Ph.D. in aerospace systems from MIT.

Previously, he was an engineering program manager on the F/A18 aircraft program at McDonnell Douglas.

Course Outline

Module 0

Logistics and Orientation

1. Instructor Intros
2. Course Structure & Master Diagram
3. Participant Intros and Expectations
4. Resources & Downloads
5. Master Glossary (inc. for Download)

Module 1

Industry Platforms: The Basics

1. An Introduction to Industry Platforms
2. Network Effects
3. The Price of Network Effects: Opening Up a Second Side

Module 2

What Does a Successful Digital Ecosystem Look Like?

1. Successful Case Studies and Analysis
2. The Five Questions
3. Case studies for Failure
4. Research on Success Rate

Module 3

Initiation and First Transactions

1. Industry Platform Essentials
2. Pricing the Platform
3. Feeling the Marble
4. Kick-starting One Side
5. Summary

Module 4

Architecture of Digital Platforms

1. Overview
2. Technical Platforms
3. Scaling
4. Industry Standards & Building Technical Coalitions

Module 5

Growing the Platform: Ensuring Quality and Robustness

1. Gating – Establishing Quality
2. “Open” and Regulation
3. Scaling – Regulation
4. Summary

Module 6

Growing the Platform: Users and Partners

1. Lead Users & Partners
2. Broader Cosystems
3. Antitrust: Growing too Much?
4. Factors in Competition
5. Summary

Module 7

Platform Dynamics

1. How can we Model a Platform?
2. Modeling Network Effects.
3. Exercise and Case Study: Google Ara
4. Trust
5. Summary

Module 8

Summary and Recap

1. Recap and Return to Master Diagram.
2. Deep Dive into Assignment (Capstone Activity)
3. One last Success (Underdog) Case?



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Certificate of Completion

All participants who successfully complete the course will receive an MIT Professional Education Certificate of Completion. In addition, they will also earn 6.4 Continuing Education Units (MIT CEUs).

In order to obtain CEUs, participants must complete a required CEU confirmation form. CEUs are calculated based on the number of learning hours in each course.



The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



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Welcome to the gateway

to MIT knowledge
and expertise for
professionals
around the world.

155

Participants from
155 countries

≠ 1600

More than 1,600
online activities

92%

92% of Professional
Alumni rated
their experience
as outstanding

About MIT Professional Education

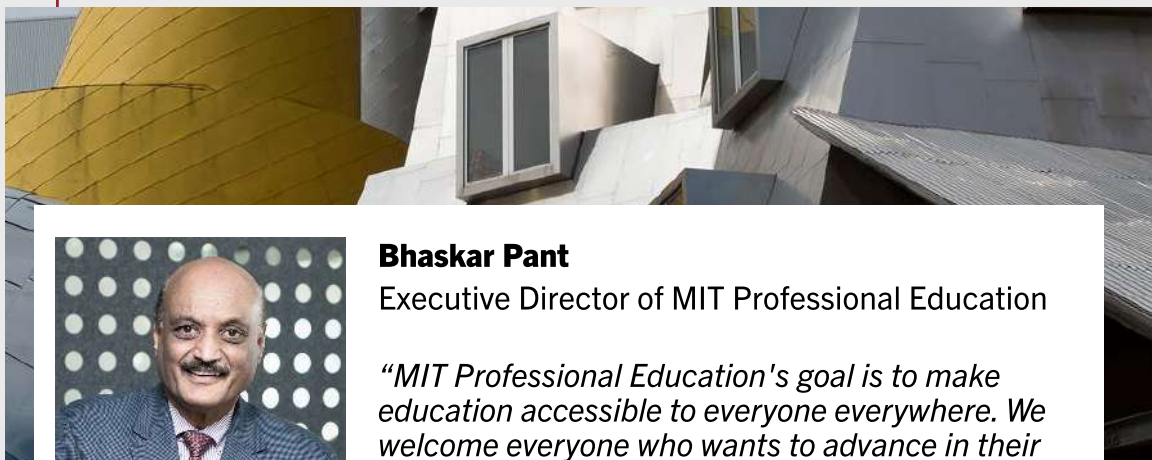
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind**.



Bhaskar Pant

Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:

Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs:

Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:

Available in Open Enrollment courses with a minimum of participants.



For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

The Beyond Online Methodology

MIT Professional Education is **revolutionizing the digital learning experience**. In fact, we are no longer talking about online-only, but rather an **interactive and collaborative learning experience** that's digitally powered: a **Beyond Online experience**.

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology. You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.



Clara Piloto

Director of Global Programs at MIT Professional Education

"As the market is undergoing changes due to optimized organizational designs catered toward potential clients, this leads to increased consumer expectations. With a two-sided market, you can create value by simplifying and accelerating transactions while offering an optimized interface. A modernized business model is necessary to keep up with accelerating customer demands."

Benefits of Joining the Alumni Community



MIT Professional Education offers **a number of benefits for participants** who successfully complete our courses:

- 1 **15% discount** for MIT Professional Education online courses and short-duration, in-person programs.
- 2 **Continued access to the virtual campus for six months after completion** of the course.
- 3 **Exclusive announcements** of new courses, programs, and events through our **virtual Alumni Campus: Professional Network**.
- 4 Invitation to the **MIT Professional Education Alumni LinkedIn group**.
- 5 **Networking opportunities** with other MIT Professional Education Alumni.



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Are you ready to develop the digital platforms that will shape the industry's future?

Frequently Asked Questions

For more information, please contact us:

+1 617 870 0270

Email: professionalprograms@mit.edu

Website: <https://professionalprograms.mit.edu>